(ii) Prove that $\text{Ker}(T/W) = \text{Ker}(T) \cap W$.

Let $x \in \text{Ker}(T/W)$

$\Rightarrow T/W(x) = 0$

$\Rightarrow T(x) = 0$

$\Rightarrow x \in \text{Ker } T$ and $x \in W$

$\therefore x \in \text{Ker}(T) \cap W$.

$\therefore \text{Ker}(T/W) \subseteq \text{Ker}(T) \cap W$.

Again, let $y \in \text{Ker}(T) \cap W$.

$\Rightarrow y \in \text{Ker } T \wedge y \in W$

$\Rightarrow y \in \text{Ker}(T/W)$

$\therefore \text{Ker}(T) \cap W \subseteq \text{Ker}(T/W)$

$\therefore \text{Ker}(T/W) = \text{Ker}(T) \cap W$.

(iii) $\text{Im}(T/W) = T(W)$

Soi:- Let any arbitary, $y \in \text{Im}(T/W)$

$\Rightarrow (T/W)(x) = y$, for some $x \in W$

$\Rightarrow T(x) = y$

$\Rightarrow y \in T(W)$

Since y is arbitary,

$\text{Im}(T/W) = T(W)$

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